Nonuniform additive constants for Lebesgue functions

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Abstract

For every $M > 0$, we construct a triangular array of interpolation nodes whose Lebesgue functions satisfy $\lambda_n(x) \le (2/\pi)\log n - M$ for all sufficiently large n at each fixed $x \in (-1,1)$. The same array admits no finite additive constant for which the sharp logarithmic lower bound holds infinitely often on a dense set. The construction uses a compact set with divergent logarithmic integrals and real-rooted polynomials with prescribed amplitudes. The results are formalized in Lean 4.

1 Introduction

For each $n \ge 1$, let

$$-1 \le x_{1,n} < \cdots < x_{n,n} \le 1,$$

and define

$$P_n(x) = \prod_{k=1}^{n}(x - x_{k,n}),$$

$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P_n'(x_{k,n})}, \qquad \lambda_n(x) = \sum_{k=1}^{n} |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. We allow arbitrary triangular arrays; the rows need not be nested.

Erdős [2, p. 68] asked whether the bound $\lambda_n(x) > (2/\pi)\log n - C$ holds infinitely often at some point, a question recorded as part of Problem 1132 [1]. The dependence allowed for C is not specified; see [4, Remark 1.12]. We show that two uniform versions fail.

Theorem 1. *For every $M > 0$, there is a triangular array of distinct nodes in $(-1,1)$ such that, for every fixed $x \in (-1,1)$,*

$$\lambda_n(x) \le \frac{2}{\pi}\log n - M \qquad \text{for all sufficiently large } n. \tag{1.1}$$

The starting index may depend on x. The array can also be chosen so that, for every finite C, the set

$$G_C = \left\{x \in (-1,1) : \lambda_n(x) > \frac{2}{\pi}\log n - C \text{ for infinitely many } n\right\} \tag{1.2}$$

is not dense in $(-1,1)$.

For a fixed array X, write $G_C(X)$ for the set in (1.2). The theorem rules out

$$\exists C\ \forall X,\quad G_C(X)\neq\varnothing,\qquad \forall X\ \exists C,\quad G_C(X) \text{ is dense.}$$

The point-dependent statement $\bigcup_{C\in\mathbb{R}} G_C(X)$ is dense for every X is proved in [3, Theorem 1(i)].

Tao [4, Theorem 1.10(i)] proved that every fixed nondegenerate interval $J\subset[-1,1]$ satisfies

$$\sup_{x\in J}\lambda_n(x)\geq \frac{2}{\pi}\log n - O_J(1).$$

Corollary 4 shows that this additive constant cannot in general be chosen independently of J.

Sections 2–4 prove Theorem 1; the final section gives the interval consequence.

2 A logarithmic integral construction

Throughout, $I=[-1,1]$; smooth on I means extendible to a smooth function on a neighborhood of I. All logarithms are natural, and $|E|$ denotes Lebesgue measure.

2.1 The compact set

For $I=[-1,1]$, write

$$Lv(x)=\int_I \frac{v(x)-v(y)}{|x-y|}\,dy.$$

In particular,

$$\|Lv\|_\infty\leq 2\|v'\|_\infty \qquad (v\in C^1(I)). \tag{2.1}$$

Fix $R>1$. Let $\varrho>0$ be sufficiently small, to be chosen below, and put

$$\delta_k=\varrho\,2^{-k},\qquad m_k=\left\lceil e^{1/\delta_k}\right\rceil.$$

At stage k, replace each retained interval by m_k equally long closed children, retaining its endpoints and removing a proportion δ_k in equal intervening gaps. The child and gap lengths are

$$\ell_k=\frac{(1-\delta_k)\ell_{k-1}}{m_k},\qquad b_k=\frac{\delta_k\ell_{k-1}}{m_k-1},\qquad \ell_0=2.$$

Let F be the intersection of the retained sets and $U=I\setminus F$. Then F is compact, contains the endpoints of I, and has empty interior. Also,

$$|U|\leq 2\sum_k \delta_k=2\varrho.$$

For an absolute $\kappa>0$, independent of ϱ, we have

$$|F\cap(x-r,x+r)|\geq\kappa r\qquad (x\in F,\ 0<r\leq 1), \tag{2.2}$$

and

$$\int_U\frac{dy}{|x-y|}=\infty\qquad (x\in F). \tag{2.3}$$

To prove (2.2), choose the first k with $\ell_k\leq r/16$, and let J be the stage $k-1$ parent containing x. At least half of every retained interval belongs to F. If $|J|\leq r$, then $J\subset[x-r,x+r]$ and

$|J| > r/16$. Otherwise $J \cap [x-r, x+r]$ has length $s \geq r$. Writing $a = \ell_k$, $b = b_k$, we have $b \leq a$ and $b/(a+b) \leq 2\delta_k$. The full children in this intersection therefore have total length at least

$$(1-2\delta_k)s - 2(a+b) \geq r/2.$$

Their intersections with F have total length at least $r/4$. This proves the density bound with an absolute constant.

For (2.3), one side of x in its stage $k-1$ parent contains at least $(m_k - 1)/2$ gaps. Among the first $\lfloor m_k/8 \rfloor$ on that side, the j-th has length at least $\delta_k \ell_{k-1}/m_k$ and lies within distance $4j\ell_{k-1}/m_k$ of x. Thus the contribution from this stage is

$$\gg \delta_k \sum_{j \leq m_k/8} \frac{1}{j} \gg \delta_k \log m_k \geq 1.$$

The gaps at different stages are disjoint, so their contributions diverge.

2.2 The logarithmic integral estimate

On each gap (b, c), put

$$r(x) = \frac{(x-b)(c-x)}{c-b},$$

and put $r = 0$ on F. Then r is 1-Lipschitz and

$$r(x) \leq \operatorname{dist}(x, F) \leq 2r(x) \qquad (x \in U).$$

Fix $B = e^8$, put $\eta = \kappa/4$, and choose $0 < \alpha < \eta$. Define

$$\phi(t) = [\log(B/t)]^{-\alpha}, \qquad u(x) = \phi(r(x)) \quad (x \in U), \qquad u = 0 \quad (x \in F). \tag{2.4}$$

The function u is continuous on I, smooth and positive in every gap, and bounded by one. We will show that there are constants $c, C > 0$ and $s_0 > 0$, independent of ϱ, such that

$$\frac{Lu(x)}{u(x)} \geq c \log \frac{1}{r(x)} - C \qquad (x \in U,\ r(x) < s_0). \tag{2.5}$$

Fix $x \in U$, write $s = r(x)$, $\Theta = \log(B/s)$, and set

$$M_I(t) = |I \cap (x-t, x+t)|, \qquad M_U(t) = |U \cap (x-t, x+t)|.$$

A point of F is within distance $2s$ of x. For $4s \leq t \leq 2$, the density bound (2.2) applied there gives at least $\kappa t/2$ of F within distance t of x. Since $M_I(t) \leq 2t$,

$$M_U(t) \leq (1-\eta) M_I(t) \qquad (4s \leq t \leq 2). \tag{2.6}$$

The contribution from $|y - x| < 4s$ to $Lu(x)$ is $O(\phi(s))$. Indeed, on $|y-x| < s/2$, the derivative formula for ϕ and Lipschitz continuity of r bound the difference quotient by $O(\phi(s)/s)$. On the remaining annulus, use $r(y) \leq 5s$ and $\phi(5s) \ll \phi(s)$. These bounds hold whenever Θ exceeds a fixed threshold.

Put $l_\pm = 1 \pm x$ and $h_\pm = \max\{\log(l_\pm/(4s)), 0\}$. The positive far contribution is $\phi(s)(h_- + h_+)$. Since $r(y) \leq s + |x-y| \leq 5|x-y|/4$ in the far region, the negative contribution is at most $\int_{4s}^2 w(t)\, dM_U(t)$, where $w(t) = \phi(5t/4)/t$ is decreasing. Integration by parts and (2.6) give

$$\int_{4s}^2 w(t)\, dM_U(t) \leq (1-\eta) \left[w(4s) M_I(4s) + \int_{4s}^2 w(t)\, dM_I(t) \right].$$

The boundary term is $O(\phi(s))$. Each side with $l_\pm > 4s$ contributes at most

$$\int_{4s}^{l_\pm} \frac{\phi(5t/4)}{t}\,dt = \int_0^{h_\pm} (\Theta - \log 5 - z)^{-\alpha}\,dz \le \frac{(\Theta - \log 5)^{-\alpha}}{1-\alpha}\,h_\pm.$$

Here $0 < h_\pm < \Theta - \log 5$, and the last inequality follows by integrating the power explicitly.

The choice $\alpha < \eta$ ensures that

$$\frac{1-\eta}{1-\alpha}\left(\frac{\Theta}{\Theta - \log 5}\right)^{\alpha} \longrightarrow \frac{1-\eta}{1-\alpha} < 1.$$

For all sufficiently large Θ, the far negative contribution is therefore at most $C\phi(s)+(1-c)\phi(s)(h_-+h_+)$, for some $c > 0$. Including the near part yields

$$\frac{Lu(x)}{u(x)} \ge c(h_- + h_+) - C \ge c\log\frac{1}{s} - C,$$

after adjusting C, since at least one of l_-, l_+ is at least one. This proves (2.5).

Now choose ϱ small enough that $\varrho/2 < s_0$ and $c\log(2/\varrho) - C > R+2$. Every gap has length at most 2ϱ, so $r(x) \le \varrho/2$. Consequently,

$$\frac{Lu(x)}{u(x)} > R + 2 \qquad (x \in U). \tag{2.7}$$

The ratio also tends to infinity as x approaches either endpoint of any fixed gap.

2.3 Smooth positive functions

Choose a smooth $\zeta : \mathbb{R} \to [0,1]$ with $\{\zeta > 0\} = U$, and a nondecreasing smooth $q_0 : \mathbb{R} \to [0,1]$, equal to zero on $(-\infty, 1]$ and one on $[2,\infty)$. For $\tau > 0$, put $\chi_\tau = q_0(\zeta/\tau)$. Its support is a compact subset of U, and $\chi_\tau \uparrow 1$ on U as $\tau \downarrow 0$.

For $x \in F$, the continuous functions

$$A_\tau(x) = \int_U \frac{\chi_\tau(y)}{|x-y|}\,dy$$

increase to infinity by (2.3) and monotone convergence. Compactness of F makes this uniform: for any threshold, the open sets on which A_τ exceeds it cover F, and a finite subcover suffices.

Fix $\epsilon > 0$. Since u is continuous and zero on F, it is at most $\epsilon/2$ in a neighborhood of F; the integral over the remainder is bounded uniformly for $x \in F$. Hence

$$\int_U \frac{\chi_\tau(y)(\epsilon - u(y))}{|x-y|}\,dy \ge \frac{\epsilon}{2}A_\tau(x) - C_\epsilon \longrightarrow \infty$$

uniformly on F. Take $\epsilon_j = 1/j$, and choose $0 < \tau_j \le 1/j$ so that this integral is at least $(R+1)\epsilon_j$ for every $x \in F$. Define

$$v_j = \chi_{\tau_j} u + (1 - \chi_{\tau_j})\epsilon_j. \tag{2.8}$$

These functions are positive and smooth on a neighborhood of I, and $Lv_j \ge (R+1)v_j$ on F.

At each fixed $x \in U$, eventually $v_j = u$ on a neighborhood of x. Bounded convergence also gives $v_j \to u$ in $L^1(I)$, so the integrals outside that neighborhood converge. Thus

$$\frac{Lv_j(x)}{v_j(x)} \longrightarrow \frac{Lu(x)}{u(x)} > R + 2 \qquad (x \in U). \tag{2.9}$$

Together with the bound on F, this gives

$$\frac{Lv_j(x)}{v_j(x)} \ge R + 1 \quad \text{eventually for each fixed } x \in I. \tag{2.10}$$

3 From amplitudes to interpolation nodes

Lemma 2. *Let* $h(z) = \sum_{r=0}^{d} h_r z^r$ *have real coefficients, no zero on the closed unit disk, and* $h(0) > 0$*. Set*

$$v(\cos\theta) = |h(e^{i\theta})|^{-1}.$$

Let T_k *be the Chebyshev polynomial of the first kind, defined by* $T_k(\cos\theta) = \cos(k\theta)$*. For* $n > d$*, define the real polynomial*

$$P_n(x) = \sum_{r=0}^{d} h_r T_{n-r}(x). \tag{3.1}$$

For all sufficiently large n*, this polynomial has exactly* n *simple roots in* $(-1, 1)$*. The corresponding Lebesgue function satisfies, on any fixed* $K \Subset (-1, 1)$*,*

$$\lambda_n(x) \le \frac{2}{\pi}\log n + 3 - \frac{Lv(x)}{\pi v(x)} \qquad (x \in K) \tag{3.2}$$

for all sufficiently large n*, with the threshold allowed to depend on* h, K*.*

Proof. The analytic logarithm of h, chosen real at zero, is real on the real diameter. Its imaginary part $\psi(\theta) = \arg h(e^{i\theta})$ thus satisfies $\psi(0) = \psi(\pi) = 0$. We have the exact identity

$$P_n(\cos\theta) = |h(e^{i\theta})| \cos(n\theta - \psi(\theta)).$$

For $n > \|\psi'\|_\infty$, the phase increases strictly from 0 to $n\pi$. The polynomial has degree exactly n, since its leading term comes from $h_0 T_n$, with $h_0 > 0$. This gives the asserted n simple roots. At their angles θ_k,

$$|P_n'(\cos\theta_k)| = |h(e^{i\theta_k})| \frac{n - \psi'(\theta_k)}{\sin\theta_k}. \tag{3.3}$$

Put $s = (n\theta - \psi(\theta))/n$, let $\theta = \Theta_n(s)$ be the inverse, and set $X_n(s) = \cos\Theta_n(s)$. The roots correspond to $s_k = (k - \frac{1}{2})\pi/n$. Moreover $X_n \to \cos$ in $C^3([0, \pi])$. At a point $x = X_n(s)$ other than a node, put $\beta = |\cos(ns)|$. Formula (3.3) gives

$$\lambda_n(x) = \frac{\beta}{nv(x)} \sum_{k=1}^{n} \frac{v(X_n(s_k))|X_n'(s_k)|}{|X_n(s) - X_n(s_k)|}. \tag{3.4}$$

Subtract the simple singularity by defining

$$R_n(s,t) = \frac{v(X_n(t))|X_n'(t)|}{|X_n(s) - X_n(t)|} - \frac{v(X_n(s))}{|s - t|}.$$

For s in a fixed compact subinterval of $(0, \pi)$, this function of t has uniformly bounded variation. Near s, write $X_n(t) - X_n(s) = (t - s)D_n(s,t)$. The function D_n stays away from zero, and

$$A_n(s,t) = \frac{v(X_n(t))|X_n'(t)|}{|D_n(s,t)|}$$

has uniformly bounded C^2 norms, with $A_n(s,s) = v(X_n(s))$. Thus $R_n(s,t) = \operatorname{sgn}(t-s)(A_n(s,t) - A_n(s,s))/(t-s)$ has uniformly bounded C^1 norms on each side of s and a bounded jump there; at $t = s$, assign either one-sided limit. Away from s, the denominator stays away from zero. If

$\Delta = \pi/n$, each quadrature cell contributes at most Δ times the variation on that cell. Summing gives

$$\left| \Delta \sum_{k=1}^{n} R_n(s, s_k) - \int_0^\pi R_n(s,t)\,dt \right| \le \Delta \operatorname{Var}_{[0,\pi]} R_n(s,\cdot) = O(1/n).$$

This bound does not depend on the distance from s to a quadrature point.

Changing variables $y = X_n(t)$ in the integral, and first deleting $(s-\epsilon, s+\epsilon)$, gives

$$\begin{aligned} \int_0^\pi R_n(s,t)\,dt &= -Lv(x) + v(x) \log \frac{1-x^2}{X_n'(s)^2 s(\pi - s)} \\ &= -Lv(x) + v(x)\left[2\log(1-\psi'(\theta)/n) - \log(s(\pi-s))\right]. \end{aligned} \tag{3.5}$$

The cancellation follows because the deleted distances in the y variable are $|X_n'(s)|\epsilon + O(\epsilon^2)$ on both sides.

Write

$$\tau = \frac{ns}{\pi} + \frac{1}{2} = m + t, \qquad m = \lfloor \tau \rfloor, \quad 0 < t < 1.$$

Since $(s - s_k)n/\pi = \tau - k$, midpoint quadrature in (3.4) gives explicitly

$$\lambda_n(x) = \frac{\beta}{\pi}\left[\sum_{k=1}^{n} \frac{1}{|\tau - k|} + \frac{1}{v(x)} \int_0^\pi R_n(s,t')\,dt'\right] + O(1/n).$$

On the compact interval under consideration, $1 \le m \le n-1$ for large n, and $\beta = \sin(\pi t)$. Splitting the elementary harmonic sum at m gives

$$\sum_{k=1}^{n} \frac{1}{|\tau - k|} \le \log(m(n-m)) + 2 + \frac{1}{t} + \frac{1}{1-t}.$$

Also, uniformly in this compact interval,

$$\log(m(n-m)) - \log(s(\pi - s)) = 2\log n - 2\log\pi + O(1/n).$$

Substituting these estimates and (3.5) into (3.4), and using $\beta/t \le \pi$, $\beta/(1-t) \le \pi$, yields

$$\lambda_n(x) \le \beta\left[\frac{2}{\pi}\log n - \frac{Lv(x)}{\pi v(x)} + \frac{2 - 2\log\pi}{\pi}\right] + 2 + O(1/n).$$

For large n the bracket is nonnegative, uniformly on K. We may replace $\beta \le 1$ by one. Since $2 + (2 - 2\log\pi)/\pi < 3$, this proves (3.2) away from the nodes. At the nodes, $\lambda_n = 1$, and (3.2) follows for large n as well. □

Lemma 3 (Amplitude approximation)**.** *Every positive $v \in C^\infty(I)$ can be approximated in $C^1(I)$ by functions*

$$\widetilde{v}(x) = |h(e^{i \arccos x})|^{-1},$$

where h is a real polynomial with no zero on the closed unit disk and $h(0) > 0$. Moreover, $L\widetilde{v}/\widetilde{v}$ converges uniformly to Lv/v.

Proof. Approximate $1/v$ by a strictly positive real polynomial q of degree d, and form

$$J(z) = z^d q\big((z + z^{-1})/2\big).$$

This is a real polynomial, and $|J(e^{i\theta})| = q(\cos\theta) > 0$. In its factorization, replace every factor $z - r$ with $|r| < 1$ by $1 - \bar{r}z$, retaining the leading scalar. The resulting polynomial h has no zero on the closed disk and the same modulus as J on the unit circle. Conjugate roots are reflected together, so its coefficients are real; changing its sign makes $h(0) > 0$. Thus

$$|h(e^{i\theta})| = q(\cos\theta), \qquad |h(e^{i\theta})|^{-1} = q(\cos\theta)^{-1}.$$

Polynomial approximation in C^1 follows by uniformly approximating the derivative and then integrating. Inversion is continuous in C^1 on functions with a positive lower bound. Hence q^{-1} approximates v in $C^1(I)$, and (2.1) gives arbitrarily accurate uniform approximation of Lv/v. □

4 Assembly of the array

Proof of Theorem 1. Fix $M > 0$, and carry out Subsections 2.1–2.3 with $R = \pi(M + 4)$. By Lemma 3, for each $j \geq 2$ choose an amplitude $\widetilde{v}_j = |h_j|^{-1}$ so close to v_j that

$$\left\| \frac{L\widetilde{v}_j}{\widetilde{v}_j} - \frac{Lv_j}{v_j} \right\|_\infty < 1/2. \tag{4.1}$$

Let $I_j = [-1 + 1/j, 1 - 1/j]$. Choose a strictly increasing sequence of integers N_j, each larger than $\deg h_j$ and the threshold in Lemma 2 for h_j, I_j. For $N_j \leq n < N_{j+1}$, use as row n the n roots of (3.1) with $h = h_j$. Fill the finitely many earlier rows with Chebyshev roots.

Fix $x \in (-1, 1)$. The block index j tends to infinity with n, and $x \in I_j$ for all large j. Equations (2.10) and (4.1) give $L\widetilde{v}_j(x)/\widetilde{v}_j(x) \geq R + 1/2$ eventually. Hence (3.2) gives, for all sufficiently large n,

$$\lambda_n(x) \leq \frac{2}{\pi}\log n + 3 - \frac{R + 1/2}{\pi} < \frac{2}{\pi}\log n - M.$$

This proves (1.1).

For the final assertion, (2.9), (3.2), and (4.1) give at every fixed $x \in U$

$$\limsup_{n\to\infty} \left(\lambda_n(x) - \frac{2}{\pi}\log n \right) \leq 3 + \frac{1}{2\pi} - \frac{Lu(x)}{\pi u(x)}. \tag{4.2}$$

Choose a gap (b, c) with both endpoints in $(-1, 1)$. By (2.5), $Lu(x)/u(x) \to +\infty$ as $x \downarrow b$. Given any finite C, there is a nonempty interval $(b, b + \sigma)$ on which the right side of (4.2) is strictly less than $-C - 1$. This interval is disjoint from G_C, proving the second assertion. □

5 Interval constants

Corollary 4 (No uniform interval constant)**.** *For the array in the second assertion of Theorem 1, there is no finite C such that*

$$\sup_{x\in I} \lambda_n(x) \geq \frac{2}{\pi}\log n - C$$

for every nondegenerate compact interval $I \subset (-1, 1)$ and all sufficiently large n, with the threshold allowed to depend on I.

Proof. Suppose that such a C exists. If G_{C+1} misses a nondegenerate compact interval $K_0 \subset (-1, 1)$, the closed sets

$$F_N = \{x \in K_0 : \lambda_n(x) \leq (2/\pi)\log n - C - 1 \text{ for all } n \geq N\}$$

cover K_0. Baire's theorem gives an F_N containing a nondegenerate compact subinterval I. The resulting upper bound on I contradicts the assumed lower bound for large n. Thus G_{C+1} is dense, contrary to Theorem 1. $\square$

Code availability

The Lean 4 formalization is available at `https://github.com/FireflySentinel/erdos-1132`.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the counterexample construction, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

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